┗(..:: Network Fundamentals Handbook ::..)┛

🏳 What is Networking?

Networking is the concept of connecting devices together to share resources and communicate. Networks exist in various forms:

- **Physical Networks**: Public transportation systems, power grids
- **Social Networks**: Friendship circles, neighborhood communities
- **Computer Networks**: Connected technological devices

In computing, networks can range from simple connections between two devices to massive systems connecting billions of devices worldwide. These networks form the backbone of modern technology infrastructure, supporting everything from personal communications to critical infrastructure.

🏳 What is the Internet?

The Internet is a global network of networks - essentially a network that connects countless smaller networks together. Using our analogy from earlier:

- Alice, Bob, and Jim form one network
- Zayn and Toby form another network
- Alice acts as the bridge between these networks, allowing all five people to communicate

The Internet works similarly, with various connection points linking different networks together.

💣 Historical Context:

- ARPANET (late 1960s): First documented computer network, funded by the US Defense Department
- World Wide Web (**1989**): Invented by **Tim Berners-Lee**, transforming the Internet into the information-sharing platform we know today

💣 Types of Networks:

- **Private Networks**: Smaller networks within homes or organizations
- **Public Networks**: The interconnections between private networks (the Internet)

🏳 Identifying Devices on a Network

For devices to communicate effectively, they need unique identifiers, similar to how humans have:

- Names (changeable)

Sayan Biswas

- Fingerprints (permanent)

In networking, devices have:

- IP Addresses (changeable)
- MAC Addresses (permanent)

☠ IP Addresses

IP addresses are numerical labels assigned to devices on a network, consisting of four octets (IPv4) like 192.168.1.77.

💣 Types of IP Addresses:

- **Private IP Address**: Identifies devices within a local network (e.g., 192.168.1.77)
- **Public IP Address**: Identifies your network on the Internet, typically assigned by your ISP

💣 IP Address Versions:

- **IPv4**: Uses 32-bit addressing (4.29 billion possible addresses)
- **IPv6**: Uses 128-bit addressing (340 trillion+ addresses) to solve the IPv4 address shortage

☠ MAC Addresses

- Physical, hardware-level address assigned to network interfaces during manufacturing
- 12-character hexadecimal number (e.g., a4:c3:f0:85:ac:2d)
- First 6 characters identify the manufacturer
- Last 6 characters are a unique identifier
- Unlike IP addresses, MAC addresses are meant to be permanent, but they can be "spoofed"

💧 Example: MAC Address Spoofing

In the hotel WiFi scenario:

- The hotel uses MAC address filtering to control access to their paid WiFi service
- Alice has paid for WiFi, so her device's MAC address (00:12:32:2F:33:39) is on the allowed list
- Bob hasn't paid, so his device's MAC address (04:9E:44:99:A3:12) is blocked
- The router is checking each device's MAC address before allowing Internet access

The hotel Wi-Fi (or ISP system) only allows the **registered device's MAC address** to access the internet. These are not the router's MAC address. These are the MAC addresses of the client devices (Alice's and Bob's devices). The router is using these addresses to identify which devices have paid for service.

Sagar Biswas

In this exercise, Bob could potentially "spoof" (fake) his MAC address to match Alice's, tricking the router into thinking his device is Alice's device that has already paid for access. This demonstrates one of the security weaknesses of simple MAC address filtering systems.

This type of MAC spoofing is often used to demonstrate how MAC address filtering alone is not a strong security measure, as MAC addresses can be easily changed in most operating systems.

🏳 Ping (ICMP)

Ping is one of the most fundamental network diagnostic **tools** available to networking professionals. It uses **ICMP** (Internet Control Message Protocol) packets to determine the performance of a connection between devices.

💣 How Ping Works:

- Sends ICMP echo request packets to target devices.
- Waits for ICMP echo reply packets from the target.
- Measures the round-trip time between sending the request and receiving the reply
- Reports statistics including packet loss and timing information.

💧 Ping Usage:

- **Basic Syntax**: ping [destination]
- **Example**: ping 8.8.8.8 or ping google.com
- **Output Information**:
  - Number of packets sent and received
  - Percentage of packet loss
  - Round-trip time statistics (minimum, maximum, average)

```
C:\Windows\System32>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=30ms TTL=114
Reply from 8.8.8.8: bytes=32 time=31ms TTL=114
Reply from 8.8.8.8: bytes=32 time=30ms TTL=114
Reply from 8.8.8.8: bytes=32 time=32ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 30ms, Maximum = 32ms, Average = 30ms
```

💧 Practical Applications:

- Testing if a remote host is reachable
- Measuring connection quality (latency and packet loss)
- Basic network troubleshooting

Sagar Biswas

- Identifying network congestion

🏳 Introduction to Local Area Networks (LAN)

💣 What is a LAN?

A Local Area Network (LAN) is a **group of computers and devices** that are connected together in a small, specific geographic area—like a single home, office building, school, or campus. Think of it as the digital "neighborhood" where your devices live and talk to each other.

Key Idea: The "Local" in LAN means everything is connected over a short distance, allowing for fast, private, and reliable communication.

💣 Core Characteristics of a LAN

1. Limited Geographic Scope

- **Scope:** Confined to a single building or a group of nearby buildings (e.g., a campus).
- **Ownership:** Privately owned, controlled, and managed by a single organization or even an individual. This is your network.

2. High Data Transfer Rates & Low Latency

- **Speed:** LANs offer **very high** data transfer rates because the distances are short. This means fast file sharing, quick access to network resources, and smooth video streaming.
- **Modern Speeds:** Common speeds today range from **1 Gigabit per second (Gbps)** to **10 Gbps** for wired connections, and fast Wi-Fi standards like Wi-Fi 6.

3. Common Network Topologies (The "Shape" of the Network)

A topology defines how devices are physically or logically arranged.

- **Star Topology (Most Common):** All devices connect to a central device, like a network switch. If one cable fails, only that device is affected. This is the **standard** for modern Ethernet and Wi-Fi networks.
- **Bus Topology (Legacy):** All devices daisy-chain (one after another in a line) along a single central cable (the "bus"). A break in the main cable takes down the entire network. Rarely used today.

  Device1 —— Device2 —— Device3 —— Device4

- **Ring Topology (Each device has exactly two neighbors):** Devices are connected in a circular loop. Data travels around the ring in one direction. Less common; used in some specific technologies like Token Ring (historical) or Fiber Distributed Data Interface (FDDI).

  Device1 → Device2 → Device3 → Device4 → Device1

Sagar Biswas

4. Key LAN Technologies

- **Ethernet:** The absolute dominant technology for *wired* LANs. It defines the rules for communication over cables (like twisted-pair copper(A **cable made of 8 small copper wires** inside) cables with RJ-45 connectors).
- **Wi-Fi (IEEE 802.11):** The standard technology for *wireless* LANs (WLANs). It uses radio waves to connect devices to the network through an access point.

5. Essential LAN Hardware Components

- **Switch:** The intelligent heart of a modern wired LAN. It learns which devices are on which ports and efficiently forwards data *only* to the intended recipient device. Example: **Netgear GS108** (home/small office switch)
- **Router (The Gateway):** Connects your LAN to other networks, most importantly the internet. It routes traffic between your local network and the wider world. Example: If your computer wants to visit a website, the router sends your request out to the internet and brings the response back. Device: **TP-Link Archer AX50** (home Wi-Fi router)
- **Wireless Access Point (WAP):** Plugs into a switch and creates a wireless LAN, allowing Wi-Fi devices like phones and laptops to connect. Example: **Built-in** WAP in Most Home Routers
- **Network Interface Card (NIC):** The hardware in your device (PC, printer, etc.) that lets it connect to the network, either via an Ethernet port or a wireless antenna. Example: Built-in **Ethernet port** on a desktop PC or laptop, **Intel Wi-Fi 6 AX200** wireless NIC in laptops
- **End Devices:** The computers, printers, servers, phones, and IoT devices that use the network.

6. Network Addressing: How Devices Find Each Other

- **MAC Address:** A **physical address** permanently burned into every network card. It's like a device's unique Social Security Number—it doesn't change. Switches use MAC addresses to communicate *within* the LAN.
- **IP Address:** A **logical address** assigned to a device so it can participate on a TCP/IP network. It's like your device's mailing address on the network. For LANs, these are typically from private IP ranges (e.g., 192.168.x.x).
- **DHCP (Dynamic Host Configuration Protocol):** A service (often running on a router or server) that *automatically* assigns IP addresses to devices when they join the network, making management easy.

7. Fundamental LAN Security Practices

- **Firewall:** A security system (often on the router) that monitors and controls incoming and outgoing network traffic based on security rules.
- **VLANs (Virtual LANs):** A method to *logically* segment a single physical network into multiple broadcast domains. This improves security and performance by isolating groups of devices (e.g., Finance Dept. vs. Guest Wi-Fi).
- **Wi-Fi Encryption: WPA2** or **WPA3** are essential to scramble data sent over the air so unauthorized users can't understand it.

Sagar Biswas

- **MAC Address Filtering:** Allows you to create a whitelist of approved devices that can connect to your network (though not foolproof).

Summary Box

Feature	Description	Common Examples
Scope	Small geographic area	Home, office, school
Speed	High data transfer rates	1 Gbps, 10 Gbps Ethernet
Topology	Physical layout of devices	Star (most common), Bus, Ring
Technology	Rules for communication	Ethernet (wired), Wi-Fi (wireless)
Key Devices	Hardware that builds the LAN	Switch, Router, Access Point
Addressing	How devices are identified	MAC Address (physical), IP Address (logical)
Security	Protecting the network	Firewalls, VLANs, WPA3 Encryption

Why it Matters: Understanding LANs is the first step to understanding all larger networks. The internet is essentially a network of countless interconnected LANs.

---------- X ----------

Sagar Biswas